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Roll No

EE/EX-5005(1)-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Energy Conservation and Management

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Describe Types of energy audit and energy management.
b) What are the methods for maximizing system efficiency and how optimize for input energy requirements.
2. a) Briefly describe the various instruments used for energy auditing.
b) Explain the second law of thermodynamics and discuss energy conservation in thermal system.
3. a) Discuss predictive and preventive maintenance from energy conservation point of view.
b) Explain waste heat techniques and Thermal insulation in Thermal system.
4. a) How a load factor of utilities improves the energy conservation?
b) Explain payback periods with examples.

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5.
 - a) Describe economic analysis depreciation method.
 - b) How power factor improved in power system and what are advantages by P.F. improvements.

6.
 - a) How energy efficient motors are different than normal motor?
 - b) A single phase 230V, 50Hz motor takes a supply current of 27 amperes at a power factor of 0.6 lagging. The motor p.f. has been improved to 0.9 lagging by connecting a condenser in parallel. Calculate the capacitance of the condenser required.

7.
 - a) Explain energy conservation tips in a residential building.
 - b) Explain the measure which can be adopted for energy conservation in cement industry.

8. Explain in detail :
 - a) NPV method verses IRR method
 - b) Tariffs used for electricity consumers
